

PaGaLGuY: turns to Google Cloud to power leading India education network

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ABOUT PaGaLGuY

With Google Cloud, PaGaLGuY has powered more personalised and engaging experiences for its users to drive growth and increase revenue.



About PaGaLGuY

PaGaLGuY is one of India's largest online education networks with a website and an app that users can download to their Android or iOS devices.

Industries: Media & Entertainment

Location: India

Cloud SQL (<https://cloud.google.com/sql/>),

Google
Cloud
results

Supply
demand

- Supported growth to 1.5 million page views per day and demand spikes that see requests increase from about 90 per second to about 1,200 per second
- Reduced API latency from about 1 second to about 40 milliseconds
- Reduced system administration time from three to four days per week to 30 minutes every two weeks

of 1,2
request
second
latency
40 m



About Searce

Searce is an Indian-headquartered business that creates engaging software products, improves business processes and delivers high-performance outcomes for clients.



App Engine (<https://cloud.google.com/appengine/>)

ABOUT SEARCE

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Searce is an Indian-headquartered

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

and delivers high-performance outcomes

Cloud SQL (<https://cloud.google.com/sql/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Founded in 2006,

PaGaLGuY

(<https://www.pagal-guy.com/>)

started as a
forum that
enabled

students,
typically aged
between 20 and
30 to discuss
and seek advice
on academic
issues. By 2011,
PaGaLGuY had
increased its
traffic to about
250,000 page
views per month.

The business is
now one of
India's largest
education
networks and
provides an app
that users can
download to their
Android and iOS
devices.

Over the past six years, PaGaLGuY has extended its
service to include video advice from experts on
education topics and grown the number of views of
its pages to 1.5 million per day. As Head of
Technology for the business, Sandeep Kalidindi has
played a key role in ensuring PaGaLGuY is as
engaging as possible to users. "Because the product
is advertising based, the greater the user
engagement, the greater the advertising revenue,"
Kalidindi explains.

In 2015, PaGaLGuY's senior management team decided to deliver an even more relevant experience for users of the education network. "The core thing we had to do was personalise the experience for each and every student that visited PaGaLGuY," Kalidindi says. "So we had to capture each student's data to customise what they see when they open the site."

The business also found traffic to the network was straining its infrastructure. During demand peaks, created by exams involving as many as 5 million students, PaGaLGuY would be inaccessible for periods of 30 minutes to one hour. Furthermore, average API latency had climbed to an unacceptable 1 second, compromising performance.

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*—Sandeep Kalidindi, Head
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PaGaLGuY needed to access extensive compute resources to undertake its planned change. Had the business relied on a physical technology architecture to undertake the transformation, it would have had to purchase capacity equivalent to 16 new servers. “There was no way with a small team we could grow to that extent in a short time,” Kalidindi says. “This was the right moment for us to explore cloud services.”

The business established two primary requirements the selected cloud service needed to meet. First, PaGaLGuY had to be able to scale the platform with costs rising only in proportion to the increase in resources consumed. Accordingly, the business would have to minimise the number of employees required to manage the cloud environment. Second, the platform had to give PaGaLGuY easy access to student data and the ability to undertake prompt, granular analysis.

PaGaLGuY reviewed available public cloud services and determined that Google Cloud (<https://cloud.google.com/>) was the best fit for its business. “Google Cloud was considerably more mature than the alternatives, with a high degree of automation and a suite of managed services,” Kalidindi says. PaGaLGuY management then discussed with Google how to optimise cost,

performance and availability of its personalised education network on Google Cloud.

“Thanks to Google Cloud, we can easily support demand peaks that see requests per second rise from an average 90 per second to about 1,200 per second for as long as 45 minutes.”

—Sandeep Kalidindi, Head of Technology, PaGaLGuY

With assistance from Google and business transformation specialists [Searce](https://www.searce.com/) (<https://www.searce.com/>), PaGaLGuY was able to deliver the platform into production on Google Cloud in 10 months. “Searce was very proactive in ensuring the environment met our needs and

allowing us to gain priority access to Google services in development,” Kalidindi says. “Their team was integral to the success of the migration.”

PaGaLGuY has been running in production in Google Cloud for two years. The education network’s Google Cloud architecture comprises a scalable back-end built on Google App Engine

(<https://cloud.google.com/appengine/>); a managed environment for its containerised applications in Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>); messaging-oriented middleware through Google Cloud Pub/Sub

(<https://cloud.google.com/pubsub/docs/overview>); a relational database in Google Cloud SQL

(<https://cloud.google.com/sql/docs/>); a managed data analytics warehouse running in Google BigQuery

(<https://cloud.google.com/bigquery/>); stream and batch data processing through Google Cloud Dataflow

(<https://cloud.google.com/dataflow/>); and object storage in Google Cloud Storage

(<https://cloud.google.com/storage/>).

PaGaLGuY has leveraged Google Cloud services to break down its platform application from a monolithic build to a series of microservices running in Google App Engine that enable independent deployment cycles, minimise test and quality assurance overheads and provide clearer monitoring and logging.

“Once our team was exposed to Google Cloud and understood the superiority of the platform, our mindset changed from ‘let us do everything on our own’ to ‘let us do what we do best’ and delegate the remainder.”

—Sandeep Kalidindi, Head of Technology, PaGaLGuY

Running on Google Cloud has enabled PaGaLGuY to add new personalisation features and grow fourfold without having to add any new engineers or administrators to accommodate the increased traffic. The business has also used the platform to seamlessly collect and aggregate students’ data for analysis, reporting and delivering a more targeted user experience. Furthermore, PaGaLGuY has been

able to provide its management team with direct access to Google BigQuery to scrutinise data rather than require them to wait at least a day to view reports created by the product or technology teams.

Support demand peaks of 1,200 requests per second

“Thanks to Google Cloud, we can easily support demand peaks that see requests per second rise from an average 90 per second to about 1,200 per second for as long as 45 minutes,” Kalidindi says. Due to Google Cloud’s scalability, PaGaLGuY can ensure its education network remains available and performance remains consistent during those periods.

Latency cut to 40 milliseconds

The business has also reduced average API latency from 1 second to about 40 milliseconds.

Furthermore, using Google Cloud has enabled PaGaLGuY to automate most of its processes and reduce system administration requirements from three to four days a week across its team members to about half an hour per week.

The performance of Google Cloud has transformed PaGaLGuY’s culture and processes. “Once our team was exposed to Google Cloud and understood the superiority of the platform, our mindset changed from ‘let us do everything on our own’ to ‘let us do what we do best’ and delegate the remainder,” Kalidindi says. The quality of the service provided by Google Cloud means PaGaLGuY effectively

considers the cloud provider as part of its team. “We are always eager to see what new services are being launched and are extremely excited about what Google Cloud can provide as part of its roadmap.” he concludes.